

UNIT 1: INTRODUCTION TO STATISTICS & DESCRIPTIVE MEASURES

*Comprehensive Academic Notes & Extensive Problem
Bank*

Course Material: Comprehensive Business & Engineering Statistics
Deep-Dive Explanations & Solutions for Over 65 Comprehensive Case Problems
Designed for Higher Academic Reference and Exhaustive Self-Study

Generated for Higher Learning Curriculum

1. Definition, Importance of Statistics & Presentation of Data

1.1 Conceptual Foundations and Definitions

The term **Statistics** carries two distinct meanings depending on whether it is used in a plural or singular context. In the *plural sense*, statistics refers to a collection of numerical data or facts, such as records of global temperatures, national employment figures, or industrial failure rates. In the *singular sense*, statistics represents the entire scientific discipline of collecting, organizing, summarizing, presenting, analyzing, and interpreting numerical information to guide rational decision-making under conditions of uncertainty.

Historically derived from structural references such as the Latin word '*Status*' or the German term '*Statistik*', the field has evolved from simple state record-keeping into a fundamental computational science. In modern analytical frameworks, statistics is categorized into two major areas:

- **Descriptive Statistics:** Methods focused on organizing, summarizing, and visually presenting data patterns using metrics like central averages, standard deviations, and structural frequency counts.
- **Inferential Statistics:** Methods that involve taking sample data from a population and applying probability theory to make predictions, generalize findings, or test hypotheses regarding the wider population.

1.2 Structural Stages of Statistical Methodology

A complete statistical study follows a rigorous five-step pipeline. Skipping or mismanaging any single phase compromises the integrity of the entire analysis:

1. **Collection of Data:** The baseline gathering phase where raw facts are collected using techniques like direct census surveys, questionnaires, or automated database logging.
2. **Organization of Data:** Sorting, filtering, and cleaning the gathered data. This phase resolves errors, handles missing values, and groups the information logically.
3. **Presentation of Data:** Condensing raw, complex data into highly structured formats like tables, charts, and graphs to reveal underlying trends.
4. **Analysis of Data:** Applying mathematical formulas to compute metrics such as means, medians, variations, and coefficients of skewness to quantify data behavior.
5. **Interpretation of Data:** Extracting actionable insights from the mathematical results to form conclusions, make forecasts, or recommend paths for business and engineering decisions.

1.3 Academic and Industrial Importance of Statistics

Statistics is essential to modern research, governance, and industry due to its broad applications:

- **Industrial Quality Engineering:** Companies use Statistical Process Control (SPC) and metrics like the Coefficient of Variation to monitor assembly lines, minimize component defects, and ensure product reliability.
- **Economics and Forecasting:** Economists rely on data modeling to track inflation, project employment patterns, and build econometric models that shape national fiscal policies.
- **Business Management and Operations:** Businesses use statistical metrics to guide resource allocation, analyze marketing trends, optimize supply chains, and mitigate financial risks.
- **Scientific Frameworks:** Medical trials use significance testing to verify vaccine efficacy, while engineering models evaluate physical structures by analyzing stress points under varying loads.

1.4 Presentation of Data: Principles of Tabulation

Raw data is often a chaotic list of numbers. Tabulation organizes this information into a logical layout of rows and columns. A professional statistical table requires several core components:

- **Table Number:** A distinct reference identifier for indexing and citation.
- **Title:** A clear description of what data is being presented, including its scope and timeline.
- **Stubs:** The horizontal row headings that categorize the data points.
- **Captions:** The vertical column headings that define the attributes or metrics being measured.
- **Body:** The core cells containing the numerical measurements.
- **Footnotes:** Clarifications for specific cells, units of measure, or shorthand abbreviations.
- **Source Notes:** A citation tracking back to the original source to ensure validity and allow verification.

1.5 Graphical and Diagrammatic Paradigms

Visual diagrams make data trends immediately recognizable. Choosing the right chart type is essential for clear communication:

- **Bar Diagrams (Simple, Multiple, Component):** Simple bars plot discrete categories against a single metric. Multiple bars compare related variables side-by-side, and component bars stack sub-variables within a single bar to display total proportions.
- **Pie Charts:** Circular diagrams where the sector angles are proportional to each category's share of the total. Calculated as: $\text{Sector Angle} = (\text{Value} / \text{Total Value}) * 360$.
- **Histograms:** Continuous rectangular plots where class intervals are marked on the X-axis and continuous frequencies on the Y-axis. The area of each bar directly reflects its frequency density.

- **Frequency Polygons & Curves:** A frequency polygon connects the midpoints of the tops of a histogram's bars with straight lines. A frequency curve smooths out these lines to show an idealized view of the distribution's shape.
- **Ogives (Cumulative Frequency Curves):** "Less-Than" ogives track cumulative totals rising from lower limits, while "More-Than" ogives track totals falling from upper limits. The point where these two curves intersect on the graph projects down to the exact **Median** value on the X-axis.

2. Exhaustive Problem Set: Presentation & Tabulation

PROBLEM 2.1: ADVANCED PIE CHART SECTOR LAYOUT

A corporate infrastructure budget of 45,000,000 Rupees is allocated across five primary operational divisions: Hardware Provisioning (15,750,000), Software Licensing (11,250,000), Cloud Resources (9,000,000), Cybersecurity Systems (5,400,000), and Workforce Upskilling (3,600,000). Calculate the exact degrees required to generate a flawless analytical pie chart.

Solution:

Total Budget Base = 45,000,000 Rupees.

Formula: **Angle** = *(Component Value / Total Base) * 360*

1. Hardware: $(15,750,000 / 45,000,000) * 360 = 0.35 * 360 = 126.0 \text{ degrees}$
2. Software: $(11,250,000 / 45,000,000) * 360 = 0.25 * 360 = 90.0 \text{ degrees}$
3. Cloud Resources: $(9,000,000 / 45,000,000) * 360 = 0.20 * 360 = 72.0 \text{ degrees}$
4. Cybersecurity: $(5,400,000 / 45,000,000) * 360 = 0.12 * 360 = 43.2 \text{ degrees}$
5. Workforce: $(3,600,000 / 45,000,000) * 360 = 0.08 * 360 = 28.8 \text{ degrees}$

Verification Check: $126.0 + 90.0 + 72.0 + 43.2 + 28.8 = 360.0 \text{ degrees}$.

PROBLEM 2.2: DUAL-OGIVE BASELINE TABLE CONSTRUCTION

Transform the following raw continuous distribution of manufacturing stress failures into a complete Dual-Ogive reference table: Classes: 0-10 ($f=8$), 10-20 ($f=15$), 20-30 ($f=27$), 30-40 ($f=35$), 40-50 ($f=11$), 50-60 ($f=4$).

Solution:

Total Frequency $N = 8 + 15 + 27 + 35 + 11 + 4 = 100$.

Class	Frequency	Less Than Boundary	Less Than CF	More Than Boundary	More Than CF
0-10	8	Less than 10	8	More than 0	100
10-20	15	Less than 20	23	More than 10	92
20-30	27	Less than 30	50	More than 20	77
30-40	35	Less than 40	85	More than 30	50
40-50	11	Less than 50	96	More than 40	15
50-60	4	Less than 60	100	More than 50	4

PROBLEM 2.3: DECONSTRUCTION OF COMPLEX TABULAR PERCENTAGES

An engineering facility has 800 employees. 60% are male. There are 250 senior engineers, of which 180 are male. Construct a table to find the count of junior female engineers.

Solution:

Total Employees = 800.

Total Males = $0.60 \times 800 = 480$. Total Females = $800 - 480 = 320$.

Senior Engineers = 250. Senior Males = 180. Senior Females = $250 - 180 = 70$.

Junior Males = Total Males - Senior Males = $480 - 180 = 300$.

Junior Females = Total Females - Senior Females = $320 - 70 = 250$.

Category	Male	Female	Total
Senior Engineers	180	70	250
Junior Engineers	300	250	550
Total	480	320	800

The count of junior female engineers is exactly 250.

PROBLEM 2.4: COMPONENT BAR CHART METRIC APPORTIONMENT

An electronics firm tracks raw resource cost structures across three distinct manufacturing models.

Model X: Copper=40, Silicon=30, Plastics=10. Model Y: Copper=25, Silicon=50, Plastics=15. Model Z:

Copper=50, Silicon=20, Plastics=30. Convert this data into stacked percentages for drawing a component bar chart.

Solution:

Model X Total = $40 + 30 + 10 = 80$. Percentages: Copper=50%, Silicon=37.5%, Plastics=12.5%.

Model Y Total = $25 + 50 + 15 = 90$. Percentages: Copper=27.78%, Silicon=55.56%, Plastics=16.67%.

Model Z Total = $50 + 20 + 30 = 100$. Percentages: Copper=50%, Silicon=20%, Plastics=30%.

PROBLEM 2.5: CONTINUOUS CLASS REALIGNMENT FROM OPEN-ENDED STRUCTURES

Fix and rebuild the following non-continuous distribution into standard class structures: "Under 20" (f=5), 20-29 (f=12), 30-39 (f=25), 40-49 (f=18), "50 and above" (f=6). Assume the underlying distribution has a uniform step size of 10 and starts at 10.

Solution:

Re-align classes to continuous structural boundaries (10-20, 20-30, 30-40, etc.):

10-20 (f=5), 20-30 (f=12), 30-40 (f=25), 40-50 (f=18), 50-60 (f=6).

PROBLEM 2.6: SECTOR ANGLE DERIVATION FOR PRODUCTION LOSS ANALYSIS

A manufacturing plant records raw operational scrap data across 4 machinery types: Type A (420kg), Type B (180kg), Type C (300kg), Type D (100kg). Find the sector angles for a pie chart.

Solution:

Total weight = $420 + 180 + 300 + 100 = 1000\text{kg}$.

Type A: $(420/1000)*360 = 151.2 \text{ degrees}$

Type B: $(180/1000)*360 = 64.8 \text{ degrees}$

Type C: $(300/1000)*360 = 108.0 \text{ degrees}$

Type D: $(100/1000)*360 = 36.0 \text{ degrees}$

PROBLEM 2.7: FREQUENCY DENSITY DERIVATION FOR VARIABLE CLASS WIDTHS

When building a histogram for variable class widths, we must adjust the height of the bars using frequency density. Calculate the frequency density for this data: 0-10 (f=10), 10-15 (f=15), 15-25 (f=30), 25-50 (f=50).

Solution:

Formula: ***Frequency Density = Frequency / Class Width***

1. 0-10: Width = 10. Density = $10 / 10 = 1.0$

2. 10-15: Width = 5. Density = $15 / 5 = 3.0$

3. 15-25: Width = 10. Density = $30 / 10 = 3.0$

4. 25-50: Width = 25. Density = $50 / 25 = 2.0$

PROBLEM 2.8: MIDPOINT-DRIVEN ORIGINAL DISTRIBUTION RECOVERY

An automated monitoring log provides midpoints (\$m\$): 15, 25, 35, 45, 55 with corresponding frequencies: 4, 11, 19, 14, 2. Recover the underlying continuous class intervals.

Solution:

The step size between midpoints is $25 - 15 = 10$. The half-step width is $10 / 2 = 5$.

Subtract 5 for lower limits; add 5 for upper limits:

10-20 ($f=4$), 20-30 ($f=11$), 30-40 ($f=19$), 40-50 ($f=14$), 50-60 ($f=2$).

PROBLEM 2.9: THREE-VARIABLE DEMOGRAPHICS MATRIX CONSTRUCTION

A university has 2,000 students. 1,200 are enrolled in Tech and 800 in Arts. 1,100 are male, with 700 of those in Tech. 400 students are out-of-state, with 300 of those in Tech. Of the out-of-state Tech students, 200 are male. Construct a comprehensive cross-tabulation table.

Solution:

This problem requires mapping three dimensions: Department, Gender, and Residency.

Let's break down the metrics systematically:

Tech Total = 1200 (Male = 700, Female = $1200 - 700 = 500$).

Arts Total = 800. Total Male = 1100, so Arts Male = $1100 - 700 = 400$. Arts Female = $800 - 400 = 400$.

Tech Out-of-State = 300 (Male = 200, Female = $300 - 200 = 100$).

Tech In-State = Total Tech - Tech Out-of-State = $1200 - 300 = 900$. Tech In-State Male = $700 - 200 = 500$. Tech In-State Female = $500 - 100 = 400$.

Total Out-of-State = 400, so Arts Out-of-State = $400 - 300 = 100$. Arts Out-of-State Male details can be balanced out against total male rules.

Let's organize this structured breakdown into a clean reference table:

Department	Residency Status	Male	Female	Total Group
Technology	In-State	500	400	900
	Out-of-State	200	100	300
Arts	In-State	350	350	700
	Out-of-State	50	50	100
Total Institutional Matrix		1100	900	2000

PROBLEM 2.10: RE-ALIGNING OVERLAPPING CONTINUOUS INTERVALS

Rebuild the following overlapping dataset into a clean, continuous frequency distribution: 0-11 ($f=5$), 11-22 ($f=12$), 22-33 ($f=18$), 33-44 ($f=7$). Adjust the boundaries to create standard inclusive continuous intervals.

Solution:

The dataset is already structured as continuous classes with a uniform step size of 11. By standard convention, the upper limit is excluded from the current row and included in the next (e.g., 0 to less than 11, 11 to less than 22). No adjustment to the numbers is needed.

PROBLEM 2.11: PERCENTAGES TO COUNTS REVERSE CONVERSION

A marketing firm surveys a population and reports these percentages: Category A (22.5%), Category B (45.0%), Category C (32.5%). If the total sample size is 1,600 people, convert these percentages back into raw counts.

Solution:

Category A: $0.225 * 1600 = 360$ individuals

Category B: $0.450 * 1600 = 720$ individuals

Category C: $0.325 * 1600 = 520$ individuals

Verification Check: $360 + 720 + 520 = 1600$.

PROBLEM 2.12: CORRECTING CUMULATIVE INCONSISTENCIES

An operator records a "less-than" cumulative frequency list as: Less than 10 (4), Less than 20 (12), Less than 30 (9), Less than 40 (25). Identify the clear recording error and explain how to fix it.

Solution:

A "less-than" cumulative frequency curve must always rise or stay flat. The drop from 12 (Less than 20) down to 9 (Less than 30) is a logical error. To correct it, the simple frequency for that interval must be re-checked from the raw data logs.

PROBLEM 2.13: MULTI-CATEGORY EXPENDITURE SECTOR ANGLES

An operations team breaks down overhead costs into four areas: Power (6,200), Maintenance (4,100), Salaries (12,500), and Logistics (3,200). Calculate the sector angles for a pie chart.

Solution:

Total = $6,200 + 4,100 + 12,500 + 3,200 = 26,000$.

Power: $(6200 / 26000) * 360 = 85.85 \text{ degrees}$

Maintenance: $(4100 / 26000) * 360 = 56.77 \text{ degrees}$

Salaries: $(12500 / 26000) * 360 = 173.08 \text{ degrees}$

Logistics: $(3200 / 26000) * 360 = 44.31 \text{ degrees}$

Verification Check: $85.85 + 56.77 + 173.08 + 44.31 = 360.0 \text{ degrees}$.

PROBLEM 2.14: EXTRACTING SIMPLE FREQUENCIES FROM CUMULATIVE LOGS

An analyst receives a summary log showing cumulative data: Less than 50 (12), Less than 100 (30), Less than 150 (65), Less than 200 (80). Reconstruct the simple frequencies for each individual class interval.

Solution:

1. Interval 0-50: **12**

2. Interval 50-100: **$30 - 12 = 18$**

3. Interval 100-150: **$65 - 30 = 35$**

4. Interval 150-200: **$80 - 65 = 15$**

The simple frequencies are: 12, 18, 35, and 15.

PROBLEM 2.15: ADJUSTING NON-CONTINUOUS BOUNDS VIA CORRECTION FACTORS

A dataset uses discrete class groups with gaps: 15-19 ($f=7$), 20-24 ($f=15$), 25-29 ($f=22$), 30-34 ($f=6$). Apply a correction factor to adjust these into continuous boundaries for a histogram.

Solution:

The gap between rows is $20 - 19 = 1$. The correction factor is $1 / 2 = 0.5$.

Subtract 0.5 from the lower limits and add 0.5 to the upper limits:

14.5-19.5 ($f=7$), 19.5-24.5 ($f=15$), 24.5-29.5 ($f=22$), 29.5-34.5 ($f=6$).

PROBLEM 2.16: CROSS-TABULATION FROM NARRATIVE SURVEY DATA

A survey tracks 500 digital device users. 300 use Android and 200 use iOS. Among the Android users, 240 use mobile banking apps. Among iOS users, 180 use mobile banking apps. Reconstruct this narrative into a clean cross-tabulation table.

Solution:

Android: Banking = 240, No Banking = $300 - 240 = 60$.

iOS: Banking = 180, No Banking = $200 - 180 = 20$.

Platform	Using Banking Apps	No Banking Apps	Total
Android	240	60	300
iOS	180	20	200
Total	420	80	500

3. Measures of Central Tendency

3.1 Conceptual Foundations

A **Measure of Central Tendency** is a single summary metric that pinpoints the center or middle of a dataset. It provides a quick overview of the data by identifying its central balance point. In statistics, we use several types of averages, which are broadly grouped into two main categories:

- **Mathematical Averages:** Measures calculated using every value in the dataset. This group includes the Arithmetic Mean, Geometric Mean, and Harmonic Mean.
- **Positional Averages:** Measures determined by a value's specific position within an ordered dataset. This group includes the Median and Mode.

3.2 Arithmetic Mean (AM)

The Arithmetic Mean is the most common average. It is calculated by summing all the values in a dataset and dividing by the total number of observations. While it is highly reliable and easy to use, it has one major drawback: it is sensitive to extreme outliers, which can pull the mean away from the true center of the data.

Mathematical Formulation

- **Individual Data Series:**

$$\text{Mean } (X_bar) = (\Sigma X) / n$$

- **Discrete Frequency Distribution:**

$$X_bar = (\Sigma f * X) / N$$

Where $N = \Sigma f$.

- **Continuous Grouped Distribution (Step-Deviation Method):**

$$X_bar = A + [(\Sigma f * d') / N] * c$$

Where A is an assumed mean, m is the class midpoint, c is the uniform class width, and $d' = (m - A) / c$.

3.3 Geometric Mean (GM)

The Geometric Mean is calculated by multiplying all the values in a dataset and taking the n -th root of that product. It is the most appropriate measure when averaging ratios, percentages, index numbers, or growth rates over time (such as compound interest or population growth).

Mathematical Formulation

- **Individual Series:**

$$GM = \text{Antilog} [(\Sigma \log X) / n]$$

- **Discrete / Continuous Frequency Series:**

$$GM = \text{Antilog} [(\Sigma f * \log X) / N]$$

*(Note: For continuous distributions, substitute the class midpoint *m* for *X*).*

3.4 Harmonic Mean (HM)

The Harmonic Mean is the reciprocal of the arithmetic mean of the reciprocals of the dataset's values. It is best used when averaging rates, speeds, or prices that change relative to fixed quantities (for example, calculating an average travel speed over equal distance intervals).

Mathematical Formulation

- **Individual Series:**

$$HM = n / [\Sigma (1 / X)]$$

- **Discrete / Continuous Frequency Distribution:**

$$HM = N / [\Sigma (f / X)]$$

3.5 Median

The Median is the exact middle value of a dataset when the observations are arranged in ascending or descending order. It splits the data into two equal halves. Because it is a positional measure, the median is unaffected by extreme outliers, making it an excellent choice for describing skewed distributions like household income.

Mathematical Formulation

- **Individual Rank Ordering:**

$$\text{Position Index} = (n + 1) / 2$$

If *n* is even, the median is the average of the two middle terms.

- **Continuous Grouped Linear Interpolation:** First, locate the median class using the position metric *N* / 2. Then apply the interpolation formula:

$$\text{Median} = L + [((N / 2) - CF) / f] * c$$

Where L is the lower limit of the median class, CF is the cumulative frequency of the *preceding* class, f is the simple frequency of the median class, and c is the class width.

3.6 Mode

The Mode is the value that appears most frequently in a dataset. A distribution can have one mode (unimodal), two modes (bimodal), or no mode at all if every value occurs with the same frequency.

Mathematical Formulation for Continuous Data

Locate the modal class (the class with the highest frequency), then apply the following interpolation formula:

$$Mode = L + [(f_1 - f_0) / (2*f_1 - f_0 - f_2)] * c$$

Where L is the lower limit of the modal class, f_1 is its frequency, f_0 is the frequency of the preceding class, f_2 is the frequency of the succeeding class, and c is the class width.

Empirical Relationship: For moderately skewed distributions, the mean, median, and mode follow a reliable relationship:

$$Mode = 3 * Median - 2 * Mean$$

This formula is used to estimate the mode if the dataset is bimodal or irregular.

4. Exhaustive Problem Set: Central Tendency

PROBLEM 4.1: MEAN VIA STEP-DEVIATION METHOD

Calculate the arithmetic mean of the following continuous factory production distribution using the step-deviation method: 0-20 ($f=5$), 20-40 ($f=12$), 40-60 ($f=20$), 60-80 ($f=10$), 80-100 ($f=3$).

Solution:

Let the Assumed Mean $A = 50$ (the midpoint of the 40-60 class). The uniform class width $c = 20$.

Class	Frequency (f)	Midpoint (m)	$d = m - 50$	$d' = d / 20$	$f * d'$
0-20	5	10	-40	-2	-10
20-40	12	30	-20	-1	-12
40-60	20	50	0	0	0
60-80	10	70	20	1	10
80-100	3	90	40	2	6

Totals: $N = 50$, $\Sigma f*d' = -10 - 12 + 0 + 10 + 6 = -6$.

Apply the formula:

$$X_{\text{bar}} = A + [(\Sigma f*d') / N] * c$$

$$X_{\text{bar}} = 50 + [-6 / 50] * 20 = 50 + [-0.12] * 20 = 50 - 2.4 = 47.6 \text{ units.}$$

PROBLEM 4.2: GEOMETRIC MEAN OF MULTIPLICATIVE GROWTH ASSETS

An investment fund grows by 8%, 15%, 25%, and 40% over four consecutive years. Compute the exact geometric mean rate of compounded growth.

Solution:

Convert the percentage growth rates into growth factors: $X_1=1.08$, $X_2=1.15$, $X_3=1.25$, $X_4=1.40$.

Formula: $GM = \text{Antilog} [(\Sigma \log X) / n]$ with $n = 4$.

$$\log(1.08) = 0.03342$$

$$\log(1.15) = 0.06070$$

$$\log(1.25) = 0.09691$$

$$\log(1.40) = 0.14613$$

Sum of logs ($\Sigma \log X$) = $0.03342 + 0.06070 + 0.09691 + 0.14613 = 0.33716$.

Divide by n : $0.33716 / 4 = 0.08429$.

$$GM = \text{Antilog}(0.08429) = 1.2142.$$

The geometric mean growth factor is 1.2142, which corresponds to an average annual growth rate of **21.42%**.

PROBLEM 4.3: HARMONIC MEAN SPEEDS OVER SEGMENTED RUNS

A commuter train travels three consecutive 50-kilometer segments at uniform speeds of 60 km/h, 80 km/h, and 120 km/h respectively. Compute the true average speed over the entire 150-kilometer journey.

Solution:

Because the distance intervals are identical (exactly 50 km each leg), we use the individual Harmonic Mean formula for the speeds (60, 80, and 120):

Formula: $HM = n / \Sigma (1 / X)$ where $n = 3$.

$$\Sigma (1 / X) = (1 / 60) + (1 / 80) + (1 / 120)$$

Find a common denominator, which is 240:

$$1/60 = 4/240, 1/80 = 3/240, 1/120 = 2/240.$$

$$\text{Sum} = (4 + 3 + 2) / 240 = 9 / 240.$$

$$HM = 3 / (9 / 240) = (3 * 240) / 9 = 240 / 3 = 80.0 \text{ km/h}.$$

The true average speed is exactly 80.0 km/h.

PROBLEM 4.4: GROUPED MEDIAN WITH INTERPOLATION MECHANICS

Determine the median value for this frequency distribution: 10-20 ($f=4$), 20-30 ($f=11$), 30-40 ($f=19$), 40-50 ($f=14$), 50-60 ($f=2$).

Solution:

First, calculate the cumulative frequencies (CF) to locate the median class:

10-20: $f=4$, $CF=4$

20-30: $f=11$, $CF=15$

30-40: $f=19$, $CF=34$

40-50: $f=14$, $CF=48$

50-60: $f=2$, $CF=50$

Total sample size $N = 50$. Target rank position = $N / 2 = 25$.

Looking at the CF column, 25 falls into the **30-40** class interval (since $CF=15$ only covers up to position 15). This is our Median Class.

Variables: $L = 30$, $CF = 15$ (cumulative frequency before median class), $f = 19$, $c = 10$.

Apply the interpolation formula:

$$\text{Median} = L + [(N/2) - CF] / f] * c$$

$$\text{Median} = 30 + [(25 - 15) / 19] * 10 = 30 + [10 / 19] * 10 = 30 + 5.263 = 35.263.$$

PROBLEM 4.5: GROUPED CONTINUOUS MODE INTERPOLATION

Find the mode for this frequency distribution: 0-10 ($f=3$), 10-20 ($f=7$), 20-30 ($f=15$), 30-40 ($f=22$), 40-50 ($f=10$), 50-60 ($f=3$).

Solution:

The highest frequency is 22, which makes **30-40** our Modal Class.

Variables: $L = 30$, $f1 = 22$ (modal frequency), $f0 = 15$ (preceding frequency), $f2 = 10$ (succeeding frequency), $c = 10$.

Apply the mode formula:

$$\text{Mode} = L + [(f1 - f0) / (2*f1 - f0 - f2)] * c$$

$$\text{Mode} = 30 + [(22 - 15) / (2*22 - 15 - 10)] * 10$$

$$\text{Mode} = 30 + [7 / (44 - 25)] * 10 = 30 + [7 / 19] * 10 = 30 + 3.684 = 33.684.$$

PROBLEM 4.6: RECONSTRUCTING A MISSING FREQUENCY COUNT

The arithmetic mean of this distribution is known to be exactly 31.875. Find the missing frequency value (x): 0-10 ($f=4$), 10-20 ($f=7$), 20-30 ($f=x$), 30-40 ($f=12$), 40-50 ($f=5$).

Solution:

Find the midpoints (m) for each class: 5, 15, 25, 35, 45.

Set up the product terms ($f * m$):

$$4 * 5 = 20$$

$$7 * 15 = 105$$

$$x * 25 = 25x$$

$$12 * 35 = 420$$

$$5 * 45 = 225$$

$$\text{Sums: } \Sigma f = 28 + x, \Sigma (f * m) = 770 + 25x.$$

Set up the mean equation:

$$31.875 = (770 + 25x) / (28 + x)$$

$$31.875 * (28 + x) = 770 + 25x$$

$$892.5 + 31.875x = 770 + 25x$$

$$31.875x - 25x = 770 - 892.5$$

$$6.875x = 122.5 \implies x = 122.5 / 6.875 = 17.81.$$

Rounding to the nearest whole integer, the missing frequency is **18**.

PROBLEM 4.7: GEOMETRIC MEAN FOR DISCRETE DATA DISTRIBUTIONS

Calculate the geometric mean of the following discrete data distribution: Values (X): 8, 12, 18, 24 with corresponding frequencies (f): 3, 5, 4, 2.

Solution:

$$\text{Total frequency } N = 3 + 5 + 4 + 2 = 14.$$

Compute $\log X$ and the product terms $f * \log X$:

$$1. X=8: \log(8) = 0.9031 \implies 3 * 0.9031 = 2.7093$$

$$2. X=12: \log(12) = 1.0792 \implies 5 * 1.0792 = 5.3960$$

$$3. X=18: \log(18) = 1.2553 \implies 4 * 1.2553 = 5.0212$$

$$4. X=24: \log(24) = 1.3802 \implies 2 * 1.3802 = 2.7604$$

$$\text{Sum of products } (\Sigma f * \log X) = 2.7093 + 5.3960 + 5.0212 + 2.7604 = 15.8869.$$

$$\text{Divide by } N: 15.8869 / 14 = 1.1348.$$

$$GM = \text{Antilog}(1.1348) = 13.64.$$

PROBLEM 4.8: HARMONIC MEAN FOR DISCRETE DATA DISTRIBUTIONS

Calculate the harmonic mean for the following discrete distribution: Values (X): 5, 10, 15, 20 with frequencies (f): 2, 4, 3, 1.

Solution:

Total frequency $N = 2 + 4 + 3 + 1 = 10$.

Compute the fractional terms (f / X):

$$2 / 5 = 0.400$$

$$4 / 10 = 0.400$$

$$3 / 15 = 0.200$$

$$1 / 20 = 0.050$$

Sum of fractions ($\Sigma f / X$) = $0.400 + 0.400 + 0.200 + 0.050 = 1.050$.

Apply the formula:

$$HM = N / \Sigma (f / X) = 10 / 1.050 = 9.524.$$

PROBLEM 4.9: CORRECTING A MISCALCULATED MEAN VALUE

An operator calculates the mean of 40 observations as exactly 52.5. Later, they discover two errors: a value of 34 was misread as 43, and a value of 70 was misread as 7. Calculate the corrected mean.

Solution:

Original Sum = $40 * 52.5 = 2100$.

Adjust for the errors by subtracting the incorrect values and adding the correct ones:

Corrected Sum = $2100 - 43 + 34 - 7 + 70 = 2154$.

Corrected Mean = $2154 / 40 = 53.85$.

PROBLEM 4.10: MEDIAN ESTIMATION FROM AN OPEN-ENDED SET

Find the median for this open-ended distribution: "Under 50" (f=6), 50-100 (f=14), 100-150 (f=25), 150-200 (f=10), "Above 200" (f=5).

Solution:

Calculate the cumulative frequencies (CF):

Under 50: CF=6

50-100: CF=20

100-150: CF=45

150-200: CF=55

Above 200: CF=60

Total frequency $N = 60$. Target rank position = $N / 2 = 30$.

Position 30 falls into the **100-150** class interval. Because the median depends only on its own class and the preceding cumulative total, the open-ended outer boundaries do not affect the calculation.

Variables: $L = 100$, $CF = 20$, $f = 25$, $c = 50$.

Apply the formula:

$$\text{Median} = 100 + [(30 - 20) / 25] * 50 = 100 + [10 / 25] * 50 = 100 + 20 = 120.0.$$

PROBLEM 4.11: EMPIRICAL ESTIMATION OF THE MODE

In a skewed distribution of data points, the arithmetic mean is calculated as 44.2, and the median is found to be 41.5. Estimate the mode using the empirical relationship.

Solution:

Formula: $\text{Mode} = 3 * \text{Median} - 2 * \text{Mean}$

$$\text{Mode} = 3 * (41.5) - 2 * (44.2) = 124.5 - 88.4 = 36.1.$$

PROBLEM 4.12: WEIGHTED MEAN CALCULATIONS

An instructor weights grading categories as follows: Exams (50%), Labs (30%), Homework (20%). A student scores 75 on Exams, 88 on Labs, and 95 on Homework. Calculate their weighted average score.

Solution:

Formula: **Weighted Mean** = $\Sigma (w * X) / \Sigma w$

Product terms:

$$75 * 0.50 = 37.5$$

$$88 * 0.30 = 26.4$$

$$95 * 0.20 = 19.0$$

Sum of products = $37.5 + 26.4 + 19.0 = 82.9$.

The total weight is 1.00 (100%), so the final score is **82.9**.

PROBLEM 4.13: COMBINED MEAN FOR MERGED GROUPS

Group A contains 30 items with a mean of 62. Group B contains 40 items with a mean of 75. Calculate the combined mean when the two groups are merged.

Solution:

Formula: **Combined Mean** = $(n1 * X1 + n2 * X2) / (n1 + n2)$

$$\text{Combined Mean} = (30 * 62 + 40 * 75) / (30 + 40)$$

$$\text{Combined Mean} = (1860 + 3000) / 70 = 4860 / 70 = 69.43.$$

PROBLEM 4.14: MIDPOINT-DRIVEN MODE INTERPOLATION

Calculate the mode for this distribution using class midpoints: Midpoints: 15 (f=4), 25 (f=12), 35 (f=28), 45 (f=16), 55 (f=5).

Solution:

The step size between midpoints is 10, so the class width $c = 10$. Reconstruct the continuous intervals: 10-20 (f=4), 20-30 (f=12), 30-40 (f=28), 40-50 (f=16), 50-60 (f=5).

The highest frequency is 28, making **30-40** our Modal Class.

Variables: $L = 30$, $f1 = 28$, $f0 = 12$, $f2 = 16$, $c = 10$.

Apply the mode formula:

$$\text{Mode} = 30 + [(28 - 12) / (2 * 28 - 12 - 16)] * 10$$

$$\text{Mode} = 30 + [16 / (56 - 28)] * 10 = 30 + [16 / 28] * 10 = 30 + 5.714 = 35.714.$$

PROBLEM 4.15: MULTI-GROUP SPEED AVERAGE OVER FIXED ROUTES

A driver covers a route four times. Their speeds on the four runs are 30 mph, 40 mph, 45 mph, and 60 mph respectively. Calculate the true average speed across all runs.

Solution:

Because the distance of the route remains fixed for each run, use the Harmonic Mean formula:

$$HM = 4 / [(1/30) + (1/40) + (1/45) + (1/60)]$$

Find a common denominator, which is 360:

$$1/30 = 12/360, 1/40 = 9/360, 1/45 = 8/360, 1/60 = 6/360.$$

$$\text{Sum of fractions} = (12 + 9 + 8 + 6) / 360 = 35 / 360.$$

$$HM = (4 * 360) / 35 = 1440 / 35 = 41.14 \text{ mph}.$$

PROBLEM 4.16: BIMODAL ARRAY RECOVERY MECHANICS

A dataset has a mean of 50.2 and a mode of 48.5. Later, the data is found to be bimodal and irregular, making the original mode calculation unreliable. Re-estimate a more stable mode using the median, which is known to be 49.6.

Solution:

Apply the empirical formula to find a more robust estimate for the mode:

$$\text{Mode} = 3 * \text{Median} - 2 * \text{Mean}$$

$$\text{Mode} = 3 * (49.6) - 2 * (50.2) = 148.8 - 100.4 = 48.4.$$

5. Quartiles and Percentiles

5.1 Conceptual Definitions and Partitions

Quantiles are points that divide a sorted dataset into equal-sized subsets. The two most common types of quantiles used in descriptive statistics are **Quartiles** and **Percentiles**:

- **Quartiles (\$Q1, Q2, Q3\$)**: These three values divide an ordered dataset into four equal parts, each containing 25% of the data.
 - **First Quartile (\$Q1\$)**: The 25th percentile. Splits off the lowest 25% of the data.
 - **Second Quartile (\$Q2\$)**: The 50th percentile, which is exactly equal to the **Median**.
 - **Third Quartile (\$Q3\$)**: The 75th percentile. Splits off the highest 25% of the data.
- **Percentiles (\$P1, P2, \dots, P99\$)**: These 99 values divide an ordered dataset into 100 equal parts, each containing 1% of the observations. Percentiles are highly useful for relative scoring and standardized benchmarks, such as interpreting student test scores or evaluating growth charts.

5.2 Mathematical Formulation for Continuous Grouped Data

To find quantiles in a continuous frequency distribution, we use a two-step process similar to finding the median. First, locate the class interval that contains the target quantile using a rank position metric. Then, use linear interpolation to pinpoint the exact value.

Quartile Interpolation Formula

First, find the quartile class using the position metric: **Rank** = $(i * N) / 4$ (where $i = 1, 2, 3$).

$$Q_i = L + [((i * N) / 4) - CF] / f * c$$

Where L is the lower limit of the target quartile class, CF is the cumulative frequency of the preceding class, f is the simple frequency of the quartile class, and c is the class width.

Percentile Interpolation Formula

First, find the percentile class using the position metric: **Rank** = $(k * N) / 100$ (where k represents the target percentile from 1 to 99).

$$P_k = L + [((k * N) / 100) - CF] / f * c$$

Where L is the lower limit of the percentile class, CF is the cumulative frequency of the preceding class, f is the simple frequency of the percentile class, and c is the class width.

6. Exhaustive Problem Set: Quartiles & Percentiles

PROBLEM 6.1: COMPREHENSIVE QUANTILE ANALYSIS FROM GROUPED DATA

Given the following distribution of examination scores, calculate the lower quartile (\$Q1\$), the upper quartile (\$Q3\$), the 40th percentile (\$P40\$), and the 85th percentile (\$P85\$): 0-10 ($f=5$), 10-20 ($f=8$), 20-30 ($f=12$), 30-40 ($f=20$), 40-50 ($f=5$).

Solution:

First, build the cumulative frequency (CF) table:

0-10: $f=5$, $CF=5$

10-20: $f=8$, $CF=13$

20-30: $f=12$, $CF=25$

30-40: $f=20$, $CF=45$

40-50: $f=5$, $CF=50$

Total sample size $N = 50$.

Part 1: Calculate \$Q1\$

Target Position = $N / 4 = 50 / 4 = 12.5$.

Looking at the CF column, 12.5 falls into the **10-20** class interval (since $CF=5$ covers up to position 5, and $CF=13$ covers up to 13).

Variables: $L = 10$, $CF = 5$, $f = 8$, $c = 10$.

$$Q1 = 10 + [(12.5 - 5) / 8] * 10 = 10 + [7.5 / 8] * 10 = 10 + 9.375 = 19.375.$$

Part 2: Calculate \$Q3\$

Target Position = $(3 * N) / 4 = (3 * 50) / 4 = 37.5$.

Position 37.5 falls into the **30-40** class interval ($CF=25$ covers up to position 25, and $CF=45$ covers up to 45).

Variables: $L = 30$, $CF = 25$, $f = 20$, $c = 10$.

$$Q3 = 30 + [(37.5 - 25) / 20] * 10 = 30 + [12.5 / 20] * 10 = 30 + 6.25 = 36.25.$$

Part 3: Calculate \$P40\$

Target Position = $(40 * N) / 100 = (40 * 50) / 100 = 20$.

Position 20 falls into the **20-30** class interval.

Variables: $L = 20$, $CF = 13$, $f = 12$, $c = 10$.

$$P40 = 20 + [(20 - 13) / 12] * 10 = 20 + [7 / 12] * 10 = 20 + 5.833 = 25.833.$$

Part 4: Calculate \$P85\$

Target Position = $(85 * N) / 100 = (85 * 50) / 100 = 42.5$.

Position 42.5 falls into the **30-40** class interval.

Variables: $L = 30$, $CF = 25$, $f = 20$, $c = 10$.

$$P85 = 30 + [(42.5 - 25) / 20] * 10 = 30 + [17.5 / 20] * 10 = 30 + 8.75 = 38.75.$$

PROBLEM 6.2: POSITION INDEX FOR INDIVIDUAL DATA SETS

Find the values of \$Q1\$, \$Q3\$, and the 70th percentile (\$P70\$) for the following raw dataset representing daily customer counts: 43, 12, 25, 58, 40, 31, 19, 52, 36.

Solution:

* Step 1: Sort the data in ascending order:

12, 19, 25, 31, 36, 40, 43, 52, 58. (Total count $n = 9$).

1. **Calculate Q1:** Position Index = $(n + 1) / 4 = 10 / 4 = 2.5$.

Take the average of the 2nd and 3rd values: $(19 + 25) / 2 = 22.0$.

2. **Calculate Q3:** Position Index = $3 * (n + 1) / 4 = 30 / 4 = 7.5$.

Take the average of the 7th and 8th values: $(43 + 52) / 2 = 47.5$.

3. **Calculate P70:** Position Index = $70 * (n + 1) / 100 = 700 / 100 = 7$.

The value at the 7th position is exactly **43.0**.

PROBLEM 6.3: INTERQUARTILE RANGE EXTRACTION

Using the distribution from Problem 6.1, calculate the Interquartile Range (IQR) and the Semi-Interquartile Range (Quartile Deviation).

Solution:

From Problem 6.1, we have: $Q1 = 19.375$ and $Q3 = 36.25$.

1. **Interquartile Range (IQR):**

$$IQR = Q3 - Q1 = 36.25 - 19.375 = 16.875.$$

2. **Semi-Interquartile Range (Quartile Deviation):**

$$QD = (Q3 - Q1) / 2 = 16.875 / 2 = 8.438.$$

PROBLEM 6.4: PERCENTILE CALCULATIONS FROM A MIDPOINT-DRIVEN LOG

Calculate the 90th percentile (\$P90\$) for this distribution of sensor logs: Midpoints: 5, 15, 25, 35, 45 with frequencies: 6, 14, 20, 8, 2.

Solution:

The step size between midpoints is 10, so the class width $c = 10$. Reconstruct the continuous intervals:

0-10 ($f=6$, $CF=6$)

10-20 ($f=14$, $CF=20$)

20-30 ($f=20$, $CF=40$)

30-40 ($f=8$, $CF=48$)

40-50 ($f=2$, $CF=50$)

Total frequency $N = 50$. Target rank position = $(90 * 50) / 100 = 45$.

Position 45 falls into the **30-40** class interval (since $CF=40$ covers up to position 40, and $CF=48$ covers up to 48).

Variables: $L = 30$, $CF = 40$, $f = 8$, $c = 10$.

Apply the percentile formula:

$$P90 = 30 + [(45 - 40) / 8] * 10 = 30 + [5 / 8] * 10 = 30 + 6.25 = 36.25.$$

PROBLEM 6.5: DIRECT INTERPOLATION OF THE 10TH PERCENTILE

Using the dataset from Problem 6.4, calculate the 10th percentile (\$P10\$) to analyze the lower tail of the distribution.

Solution:

Target Rank Position = $(10 * 50) / 100 = 5$.

Looking at the CF column, position 5 falls into the very first class interval, **0-10**.

Variables: $L = 0$, $CF = 0$ (no preceding class), $f = 6$, $c = 10$.

Apply the formula:

$$P10 = 0 + [(5 - 0) / 6] * 10 = [5 / 6] * 10 = 8.333.$$

PROBLEM 6.6: FINDING THE PERCENTILE RANK OF A SPECIFIC VALUE

Using the sorted data array from Problem 6.2 (12, 19, 25, 31, 36, 40, 43, 52, 58), determine the empirical percentile rank of the value 40.

Solution:

Formula for Percentile Rank: **Rank** = *(Number of values below X / Total Number of values) * 100*

Count the numbers less than 40 in the sorted list: 12, 19, 25, 31, 36 (exactly 5 values).

Total count **n** = 9.

Percentile Rank = $(5 / 9) * 100 = 55.56\%$.

The value 40 sits at approximately the 55.56th percentile of this dataset.

PROBLEM 6.7: PERCENTILE RANGE CALCULATION (P90 - P10)

Using the values computed in Problem 6.4 (\$P90 = 36.25\$) and Problem 6.5 (\$P10 = 8.333\$), calculate the 10-90 percentile range.

Solution:

The percentile range measures the spread of the middle 80% of the data, stripping away extreme outliers at both ends:

Percentile Range = $P90 - P10 = 36.25 - 8.333 = 27.917$.

PROBLEM 6.8: DECILE TO PERCENTILE MAPPING

An analyst needs to find the 7th Decile (\$D7\$) for the continuous distribution given in Problem 6.4. Explain the relationship between deciles and percentiles, and calculate the value.

Solution:

Deciles divide data into 10 equal parts. The 7th Decile (\$D7\$) corresponds exactly to the 70th Percentile (\$P70\$).

Target Rank Position = $(70 * 50) / 100 = 35$.

Looking at the CF table from Problem 6.4, position 35 falls into the **20-30** class interval.

Variables: **L** = 20, **CF** = 20, **f** = 20, **c** = 10.

Apply the formula:

D7 = P70 = $20 + [(35 - 20) / 20] * 10 = 20 + [15 / 20] * 10 = 20 + 7.5 = 27.5$.

7. Measures of Variation (Dispersion)

7.1 Conceptual Foundations of Variation

Measures of central tendency locate the center of a dataset, but they do not tell us how spread out the values are. **Measures of Variation** (or dispersion) quantify the spread, volatility, or consistency of the data points. Understanding dispersion is essential for risk analysis and quality control, as it helps distinguish between stable, predictable processes and highly volatile ones.

7.2 Range

The Range is the simplest absolute measure of dispersion. It is the difference between the largest (maximum) value and the smallest (minimum) value in a dataset. While easy to calculate, it is highly sensitive to outliers and ignores how the rest of the data points are distributed between the two extremes.

$$\text{Range} = L - S$$

$$\text{Coefficient of Range} = (L - S) / (L + S)$$

7.3 Mean Deviation (MD)

The Mean Deviation is the arithmetic mean of the absolute deviations of the observations calculated from a central average (usually the mean or median). By using absolute values, it treats all deviations as positive distances, preventing negative and positive values from canceling each other out.

Mathematical Formulation

- **Individual Series (About the Median):**

$$MD_{\text{median}} = (\sum |X - \text{Median}|) / n$$

- **Frequency Distribution:**

$$MD = (\sum f * |X - \text{Average}|) / N$$

- **Relative Measure:**

$$\text{Coefficient of MD} = \text{Mean Deviation} / \text{Average Used}$$

7.4 Standard Deviation (SD)

The Standard Deviation (σ) is the most widely used and mathematically robust measure of variation. It measures the average distance of data points from the mean. Instead of using absolute value brackets to handle signs, it squares the deviations before averaging them, and then takes the square root of the result. The squared value of the standard deviation is known as the **Variance**.

$$\text{Variance} = \sigma^2$$

Mathematical Formulation

- **Individual Series Shortcut Method:**

$$\sigma = \text{Sqrt} [(\Sigma X^2 / n) - (X_{\text{bar}})^2]$$

- **Grouped Continuous Series (Step-Deviation Method):**

$$\sigma = \text{Sqrt} [(\Sigma f * d'^2 / N) - ((\Sigma f * d' / N))^2] * c$$

Where $d' = (m - A) / c$.

7.5 Coefficient of Variation (CV)

The Coefficient of Variation expresses the standard deviation as a percentage of the arithmetic mean. Because it is a dimensionless, relative measure, it is ideal for comparing the consistency, stability, or volatility of two or more datasets with different units or widely different means.

$$CV = (\sigma / X_{\text{bar}}) * 100$$

Rule of Interpretation: A dataset with a **lower CV** is considered more consistent, stable, and less variable. A dataset with a **higher CV** is more variable and less consistent.

8. Exhaustive Problem Set: Measures of Variation

PROBLEM 8.1: MEAN DEVIATION ABOUT THE MEDIAN

Calculate the Mean Deviation about the Median and its relative coefficient for this individual data array:
15, 8, 22, 10, 18, 5, 25.

Solution:

* Step 1: Sort the data to find the median: 5, 8, 10, 15, 18, 22, 25. (Count $n = 7$).

The median position is $(7 + 1) / 2 = 4$. The 4th value is **15**.

* Step 2: Compute the absolute deviations from the median, $|X - 15|$:

X	$ X - 15 $
5	$ 5 - 15 = 10$
8	$ 8 - 15 = 7$
10	$ 10 - 15 = 5$
15	$ 15 - 15 = 0$
18	$ 18 - 15 = 3$
22	$ 22 - 15 = 7$
25	$ 25 - 15 = 10$

Sum of absolute deviations ($\Sigma |X - 15|$) = $10 + 7 + 5 + 0 + 3 + 7 + 10 = 42$.

* Step 3: Apply the formulas:

$MD_{median} = 42 / 7 = 6.0$.

$Coefficient\ of\ MD = 6.0 / 15 = 0.40$.

PROBLEM 8.2: STANDARD DEVIATION & VARIANCE VIA SHORTCUT METHOD

Calculate the Mean, Variance, and Standard Deviation for this dataset using the shortcut method: 6, 8, 10, 12, 14.

Solution:

Count $n = 5$.

Compute the sum of values (ΣX) and the sum of squared values (ΣX^2):

$$\Sigma X = 6 + 8 + 10 + 12 + 14 = 46.$$

$$\text{Mean } (X_{\text{bar}}) = 46 / 5 = 9.2.$$

$$\Sigma X^2 = 36 + 64 + 100 + 144 + 196 = 540.$$

Apply the shortcut variance formula:

$$\text{Variance} = (\Sigma X^2 / n) - (X_{\text{bar}})^2$$

$$\text{Variance} = (540 / 5) - (9.2)^2 = 108 - 84.64 = 23.36.$$

Take the square root to find the Standard Deviation:

$$\sigma = \text{Sqrt}(23.36) = 4.833.$$

PROBLEM 8.3: COMPARING PROCESS CONSISTENCY VIA CV

Two automated assembly lines produce steel rods. Based on the performance metrics below, determine which line is more consistent in maintaining rod lengths: Line Alpha: Mean length = 150 mm, Standard Deviation = 4.5 mm.

Line Beta: Mean length = 180 mm, Standard Deviation = 5.2 mm.

Solution:

Calculate the Coefficient of Variation (CV) for both lines:

$$CV_{\text{Alpha}} = (4.5 / 150) * 100 = 0.03 * 100 = 3.0\%.$$

$$CV_{\text{Beta}} = (5.2 / 180) * 100 = 0.0289 * 100 = 2.89\%.$$

Conclusion: Line Beta has a lower Coefficient of Variation ($2.89\% < 3.0\%$), meaning its production process is slightly more consistent and shows less relative variation than Line Alpha.

PROBLEM 8.4: GROUPED STANDARD DEVIATION VIA STEP-DEVIATION METHOD

Calculate the standard deviation (σ) for this continuous factory output distribution using the step-deviation method: 0-10 ($f=4$), 10-20 ($f=8$), 20-30 ($f=15$), 30-40 ($f=10$), 40-50 ($f=3$).

Solution:

Let the Assumed Mean $A = 25$ (the midpoint of the 20-30 class). The class width $c = 10$.

Class	f	Midpoint (m)	$d' = (m-25)/10$	$f * d'$	$f * d'^2$
0-10	4	5	-2	-8	16
10-20	8	15	-1	-8	8
20-30	15	25	0	0	0
30-40	10	35	1	10	10
40-50	3	45	2	6	12

Totals: $N = 40$, $\Sigma f*d' = -8 - 8 + 0 + 10 + 6 = 0$, $\Sigma f*d'^2 = 16 + 8 + 0 + 10 + 12 = 46$.

Apply the step-deviation formula:

$$\sigma = \text{Sqrt} [(46 / 40) - (0 / 40)^2] * 10$$

$$\sigma = \text{Sqrt} [1.15 - 0] * 10 = \text{Sqrt}(1.15) * 10 = 1.0724 * 10 = 10.724 \text{ units.}$$

PROBLEM 8.5: COMBINED STANDARD DEVIATION FOR MERGED DATA

Two experimental datasets are merged into a single distribution. Calculate the combined standard deviation using the following group statistics:

Group 1: $n_1 = 40$, $X_1 = 60$, $\sigma_1 = 5$

Group 2: $n_2 = 60$, $X_2 = 70$, $\sigma_2 = 8$

Solution:

* Step 1: Calculate the Combined Mean (X_c):

$$X_c = (40 * 60 + 60 * 70) / (40 + 60) = (2400 + 4200) / 100 = 6600 / 100 = 66.0.$$

* Step 2: Calculate the deviation of each group's mean from the combined mean (d_1 and d_2):

$$d_1 = X_1 - X_c = 60 - 66 = -6 \text{ \textit{implies } } d_1^2 = 36.$$

$$d_2 = X_2 - X_c = 70 - 66 = 4 \text{ \textit{implies } } d_2^2 = 16.$$

* Step 3: Apply the combined standard deviation formula:

$$\sigma_c = \text{Sqrt} [(n_1 * (\sigma_1^2 + d_1^2) + n_2 * (\sigma_2^2 + d_2^2)) / (n_1 + n_2)]$$

$$\sigma_c = \text{Sqrt} [(40 * (25 + 36) + 60 * (64 + 16)) / 100]$$

$$\sigma_c = \text{Sqrt} [(40 * (61) + 60 * (80)) / 100] = \text{Sqrt} [(2440 + 4800) / 100]$$

$$\sigma_c = \text{Sqrt} [7240 / 100] = \text{Sqrt}(72.4) = 8.509.$$

PROBLEM 8.6: MEAN DEVIATION ABOUT THE MEAN (GROUPED DATA)

Calculate the Mean Deviation about the Arithmetic Mean for the discrete distribution given below: Values (X): 10, 15, 20, 25 with frequencies (f): 3, 7, 6, 4.

Solution:

Total frequency $N = 3 + 7 + 6 + 4 = 20$.

First, find the arithmetic mean:

$$\Sigma f \cdot X = (3 \cdot 10) + (7 \cdot 15) + (6 \cdot 20) + (4 \cdot 25) = 30 + 105 + 120 + 100 = 355.$$

$$\text{Mean} = 355 / 20 = 17.75.$$

Next, compute the absolute deviations from the mean, $|X - 17.75|$, and the product terms $f \cdot |X - 17.75|$:

X	f	$ X - 17.75 $	$f \cdot X - 17.75 $
10	3	7.75	23.25
15	7	2.75	19.25
20	6	2.25	13.50
25	4	7.25	29.00

Sum of products = $23.25 + 19.25 + 13.50 + 29.00 = 85.0$.

$$\text{MD}_{\text{mean}} = 85.0 / 20 = 4.25.$$

PROBLEM 8.7: ROOT-MEAN-SQUARE ALTERNATIVE DERIVATION

Verify the standard deviation of the raw dataset (4, 7, 9, 12) using the basic definitional formula: $\sigma = \text{Sqrt}(\Sigma(X - \bar{X})^2 / n)$.

Solution:

Count $n = 4$.

$$\Sigma X = 4 + 7 + 9 + 12 = 32 \text{ \textit{implies} Mean} = 32 / 4 = 8.0.$$

Compute the deviations and squared deviations from the mean:

1. $X=4: 4 - 8 = -4 \text{ \textit{implies} } 16$

2. $X=7: 7 - 8 = -1 \text{ \textit{implies} } 1$

3. $X=9: 9 - 8 = 1 \text{ \textit{implies} } 1$

4. $X=12: 12 - 8 = 4 \text{ \textit{implies} } 16$

Sum of squared deviations = $16 + 1 + 1 + 16 = 34$.

$$\sigma = \text{Sqrt}(34 / 4) = \text{Sqrt}(8.5) = 2.915.$$

PROBLEM 8.8: CORRECTING A MISCALCULATED VARIANCE VALUE

A sample of 20 items has a mean of 40 and a standard deviation of 5. Later, a value of 50 is found to be incorrect and is removed from the dataset without replacement. Calculate the corrected mean and standard deviation for the remaining 19 items.

Solution:

Original metrics: $n = 20$, $\text{Mean} = 40$, $\sigma = 5 \implies \sigma^2 = 25$.

Use the variance identity to find the original sum of squares:

$$25 = (\Sigma X^2 / 20) - 1600 \implies \Sigma X^2 / 20 = 1625 \implies \Sigma X^2 = 32500.$$

Original sum of values (ΣX) = $20 * 40 = 800$.

Remove the incorrect value (50):

New sum of values = $800 - 50 = 750$.

New Mean = $750 / 19 = 39.474$.

New sum of squares = $32500 - 50^2 = 32500 - 2500 = 30000$.

Calculate the new variance:

$$\text{New Variance} = (30000 / 19) - (39.474)^2 = 1578.947 - 1558.197 = 20.75.$$

New Standard Deviation = $\text{Sqrt}(20.75) = 4.555$.

9. Skewness

9.1 Conceptual Foundations and Asymmetry Mechanics

Skewness measures the asymmetry of a data distribution around its mean. If a distribution curve is perfectly balanced and symmetric, the left and right halves are mirror images of each other, and the mean, median, and mode are all equal. Skewness quantifies how much and in which direction the data drifts away from this symmetry:

- **Symmetrical Distribution:** The data is balanced. The tail lengths on both sides are equal, and the skewness metric is exactly zero ($Mean = Median = Mode$).
- **Positively Skewed (Right-Tailed):** The distribution has a long tail extending toward higher values on the right side. This indicates a concentration of data points at lower values, with a few extreme outliers pulling the mean upward ($Mode < Median < Mean$).
- **Negatively Skewed (Left-Tailed):** The distribution has a long tail extending toward lower values on the left side. This indicates a concentration of data points at higher values, with a few extreme low outliers pulling the mean downward ($Mean < Median < Mode$).

9.2 Karl Pearson's Method of Skewness

Karl Pearson's coefficient of skewness measures the distance between the mean and the mode, scaled by the standard deviation to make it a dimensionless, relative metric. This allows for direct comparisons between different datasets.

$$Sk_p = (Mean - Mode) / \sigma$$

If the mode is difficult to determine or the distribution is irregular, we use the empirical relationship ($Mode = 3 * Median - 2 * Mean$) to calculate the coefficient using the median instead:

$$Sk_p = [3 * (Mean - Median)] / \sigma$$

The values for this coefficient typically range between -3 and $+3$. A positive result indicates right-skewness, while a negative result indicates left-skewness.

9.3 Bowley's Method of Skewness

Bowley's coefficient of skewness is based entirely on positional measures (quartiles). It evaluates asymmetry by comparing the relative distances of the first ($Q1$) and third ($Q3$) quartiles from the median ($Q2$).

$$Sk_b = [(Q3 - Q2) - (Q2 - Q1)] / (Q3 - Q1) = (Q3 + Q1 - 2*Q2) / (Q3 - Q1)$$

Because the median is identical to the second quartile ($Q2$), this formula can be rewritten as:

$$Sk_b = (Q3 + Q1 - 2 * Median) / (Q3 - Q1)$$

Bowley's coefficient focuses on the behavior of the middle 50% of the dataset, making it completely unaffected by extreme outliers in the outer tails. Its values range strictly between -1 and $+1$.

10. Exhaustive Problem Set: Skewness Analysis

PROBLEM 10.1: KARL PEARSON SKEWNESS DERIVATION VIA MODE IDENTITY

An automated analysis of transaction records yields an arithmetic mean of 145.0 Rupees, a mode of 132.0 Rupees, and a standard deviation of 22.5 Rupees. Compute Karl Pearson's coefficient of skewness and interpret the shape of the distribution.

Solution:

Apply Karl Pearson's mode-based formula:

$$Sk_p = (Mean - Mode) / \sigma$$

$$Sk_p = (145.0 - 132.0) / 22.5 = 13.0 / 22.5 = +0.578.$$

****Interpretation:**** The positive coefficient ($+0.578$) indicates that the transaction distribution is ****positively skewed**** (right-tailed), with a longer tail extending toward higher values.

PROBLEM 10.2: KARL PEARSON SKEWNESS DERIVATION VIA MEDIAN IDENTITY

A survey of household internet usage shows a mean of 44.2 hours per week, a median of 40.8 hours, and a standard deviation of 12.4 hours. Calculate Karl Pearson's coefficient of skewness using the median-based formula.

Solution:

Because the mode isn't explicitly provided, use the alternative median-based formula:

$$Sk_p = [3 * (Mean - Median)] / \sigma$$

$$Sk_p = [3 * (44.2 - 40.8)] / 12.4 = [3 * 3.4] / 12.4 = 10.2 / 12.4 = +0.823.$$

****Interpretation:**** The positive result ($+0.823$) shows that the data is ****moderately right-skewed****.

PROBLEM 10.3: BOWLEY QUARTILE SKEWNESS CALCULATION

A real estate study analyzes property values and reveals the following quartile boundaries: First Quartile (\$Q1\$) = 35,000 Rupees, Median (\$Q2\$) = 58,000 Rupees, and Third Quartile (\$Q3\$) = 96,000 Rupees. Calculate Bowley's coefficient of skewness.

Solution:

Apply Bowley's quartile formula:

$$Sk_b = (Q3 + Q1 - 2 * Median) / (Q3 - Q1)$$

$$Sk_b = (96000 + 35000 - 2 * 58000) / (96000 - 35000)$$

$$Sk_b = (131000 - 116000) / 61000 = 15000 / 61000 = +0.246.$$

****Interpretation:**** The positive coefficient (\$+0.246\$) shows that the distribution of property values is ****positively skewed****. The upper 25% of properties show a wider spread in value than the lower 25%.

PROBLEM 10.4: GROUPED SKEWNESS PROFILE ANALYSIS

Calculate Karl Pearson's coefficient of skewness for the continuous frequency distribution given below:
0-10 ($f=5$), 10-20 ($f=12$), 20-30 ($f=22$), 30-40 ($f=8$), 40-50 ($f=3$).

Solution:

To find the coefficient, we need to calculate the Mean, Mode, and Standard Deviation for this distribution.
Let the Assumed Mean $A = 25$ and class width $c = 10$.

Class	f	m	d'	f*d'	f*d'^2
0-10	5	5	-2	-10	20
10-20	12	15	-1	-12	12
20-30	22	25	0	0	0
30-40	8	35	1	8	8
40-50	3	45	2	6	12

Totals: $N = 50$, $\Sigma f*d' = -8$, $\Sigma f*d'^2 = 52$.

1. **Calculate the Mean:**

$$\text{Mean} = 25 + (-8 / 50) * 10 = 25 - 1.6 = 23.4.$$

2. **Calculate the Standard Deviation (σ):**

$$\sigma = \text{Sqrt} [(52 / 50) - (-8 / 50)^2] * 10 = \text{Sqrt} [1.04 - 0.0256] * 10 = \text{Sqrt}(1.0144) * 10 = 1.0072 * 10 = 10.072.$$

3. **Calculate the Mode:**

The highest frequency is 22, making **10-20** the Modal Class.

Variables: $L = 10$, $f_1 = 22$, $f_0 = 12$, $f_2 = 8$, $c = 10$.

$$\text{Mode} = 10 + [(22 - 12) / (2*22 - 12 - 8)] * 10 = 10 + [10 / 24] * 10 = 10 + 4.167 = 14.167.$$

4. **Calculate Karl Pearson's Skewness:**

$$\text{Sk}_p = (\text{Mean} - \text{Mode}) / \sigma = (23.4 - 14.167) / 10.072 = 9.233 / 10.072 = +0.917.$$

PROBLEM 10.5: INVERSE MAPPING FROM SKEWNESS METRICS

A distribution has a Karl Pearson coefficient of skewness of -0.40, an arithmetic mean of 65.0, and a standard deviation of 15.0. Reconstruct the implied value of the mode.

Solution:

Formula: $Sk_p = (Mean - Mode) / \sigma$

Substitute the given values into the equation:

$$-0.40 = (65.0 - Mode) / 15.0$$

$$-0.40 * 15.0 = 65.0 - Mode$$

$$-6.0 = 65.0 - Mode \implies Mode = 65.0 + 6.0 = 71.0.$$

The implied mode value is **71.0**.

PROBLEM 10.6: RECONSTRUCTING THE MEDIAN FROM SKEWNESS RULES

Using the dataset from Problem 10.5, calculate the implied median value of the distribution.

Solution:

Apply the empirical relationship formula:

$$Mode = 3 * Median - 2 * Mean$$

Substitute the known values ($Mode = 71.0$, $Mean = 65.0$):

$$71.0 = 3 * Median - 2 * (65.0)$$

$$71.0 = 3 * Median - 130.0$$

$$3 * Median = 71.0 + 130.0 = 201.0$$

$$Median = 201.0 / 3 = 67.0.$$

PROBLEM 10.7: BOWLEY SKEWNESS VERIFICATION ON SYMMETRIC SETS

Prove that Bowley's coefficient of skewness equals exactly zero for a perfectly balanced, symmetric dataset where $Q1 = 20$, $Median = 35$, and $Q3 = 50$.

Solution:

Apply Bowley's formula:

$$Sk_b = (Q3 + Q1 - 2 * Median) / (Q3 - Q1)$$

Substitute the values into the numerator:

$$Numerator = 50 + 20 - 2 * (35) = 70 - 70 = 0.$$

$$Denominator = 50 - 20 = 30.$$

$$Sk_b = 0 / 30 = 0.0.$$

This confirms that a balanced distribution has zero skewness.

PROBLEM 10.8: SKEWNESS DIRECTION ANALYSIS FROM COMPONENT VARIABLES

An analyst reports these quartile measurements for a quality control dataset: $Q1 = 14.2$, Median = 22.4, $Q3 = 25.1$. Determine the direction of skewness using Bowley's method.

Solution:

Apply Bowley's quartile formula:

$$Sk_b = (Q3 + Q1 - 2 * Median) / (Q3 - Q1)$$

$$Sk_b = (25.1 + 14.2 - 2 * 22.4) / (25.1 - 14.2)$$

$$Sk_b = (39.3 - 44.8) / 10.9 = -5.5 / 10.9 = -0.505.$$

Interpretation: The negative result (-0.505) indicates that the distribution is **negatively skewed** (left-tailed).

11. Exhaustive Unit-Level Comprehensive Case Set

This final module provides advanced, integrated problems that combine concepts from across the entire unit—tabulation, central tendency, quantiles, dispersion, and skewness—to test and reinforce your comprehensive data analysis skills.

CASE PROBLEM 11.1: COMPLETE END-TO-END ANALYSIS MATRIX

An engineering team measures stress limits across 100 sample components. Reconstruct the continuous frequency distribution below and calculate its Mean, Median, Standard Deviation, and Karl Pearson Skewness:

Continuous Data: 10-20 ($f=10$), 20-30 ($f=20$), 30-40 ($f=40$), 40-50 ($f=20$), 50-60 ($f=10$).

Solution:

Let the Assumed Mean $A = 35$ (midpoint of the 30-40 class) and class width $c = 10$.

Class	f	m	d'	f*d'	f*d'^2
10-20	10	15	-2	-20	40
20-30	20	25	-1	-20	20
30-40	40	35	0	0	0
40-50	20	45	1	20	20
50-60	10	55	2	20	40

Totals: $N = 100$, $\Sigma f*d' = 0$, $\Sigma f*d'^2 = 120$.

1. **Calculate the Mean:**

$$\text{Mean} = 35 + (0 / 100) * 10 = 35.0.$$

2. **Calculate the Median:**

Target Position = $N / 2 = 50$. Looking at the cumulative frequencies (10, 30, 70), position 50 falls into the **30-40** class.

Variables: $L = 30$, $CF = 30$, $f = 40$, $c = 10$.

$$\text{Median} = 30 + [(50 - 30) / 40] * 10 = 30 + [20 / 40] * 10 = 30 + 5 = 35.0.$$

3. **Calculate the Standard Deviation (σ):**

$$\sigma = \text{Sqrt} [(120 / 100) - (0)^2] * 10 = \text{Sqrt}(1.2) * 10 = 1.0954 * 10 = 10.954.$$

4. **Calculate Karl Pearson Skewness:**

$$Sk_p = [3 * (\text{Mean} - \text{Median})] / \sigma = [3 * (35.0 - 35.0)] / 10.954 = 0.0.$$

Conclusion: Because the Mean equals the Median, the distribution is **perfectly symmetric** with a skewness coefficient of zero.

CASE PROBLEM 11.2: ADVANCED DOUBLE-MISSING FREQUENCY SYNTHESIS

A distribution has a known mean of 44.0 and a total frequency count (N) of 100. Reconstruct the two missing frequency values (x and y):

0-20 ($f=15$), 20-40 ($f=x$), 40-60 ($f=y$), 60-80 ($f=12$), 80-100 ($f=8$).

Solution:

* Step 1: Set up the total frequency equation:

$$15 + x + y + 12 + 8 = 100 \implies 35 + x + y = 100 \implies x + y = 65 \implies y = 65 - x.$$

* Step 2: Calculate the class midpoints (m): 10, 30, 50, 70, 90. Set up the product terms ($f * m$):

$$15 * 10 = 150$$

$$x * 30 = 30x$$

$$y * 50 = 50y$$

$$12 * 70 = 840$$

$$8 * 90 = 720$$

$$\text{Sum of products } (\Sigma f * m) = 150 + 30x + 50y + 840 + 720 = 1710 + 30x + 50y.$$

* Step 3: Substitute the mean identity into the equation:

$$\text{Mean} = \Sigma f * m / N \implies 44.0 = (1710 + 30x + 50y) / 100$$

$$4400 = 1710 + 30x + 50y \implies 30x + 50y = 4400 - 1710 = 2690.$$

Simplify by dividing by 10: $3x + 5y = 269$.

* Step 4: Substitute $y = 65 - x$ into the simplified equation:

$$3x + 5 * (65 - x) = 269$$

$$3x + 325 - 5x = 269$$

$$-2x = 269 - 325 = -56 \implies x = 28.$$

* Step 5: Solve for y :

$$y = 65 - 28 = 37.$$

The missing frequency counts are $x = 28$ and $y = 37$.

CASE PROBLEM 11.3: VERIFICATION OF MATHEMATICAL INEQUALITIES ($AM \geq GM \geq HM$)

Prove the mathematical rule that the Arithmetic Mean is greater than or equal to the Geometric Mean, which is greater than or equal to the Harmonic Mean ($AM \geq GM \geq HM$), using this raw dataset: 4, 8, 16.

Solution:

Count $n = 3$.

1. **Calculate the Arithmetic Mean (AM):**

$$AM = (4 + 8 + 16) / 3 = 28 / 3 = 9.333.$$

2. **Calculate the Geometric Mean (GM):**

$$GM = \text{Sqrt}[3](4 * 8 * 16) = \text{Sqrt}[3](512) = 8.0.$$

3. **Calculate the Harmonic Mean (HM):**

$$HM = 3 / [(1/4) + (1/8) + (1/16)]$$

Find a common denominator, which is 16:

$$1/4 = 4/16, 1/8 = 2/16, 1/16 = 1/16.$$

$$\text{Sum of fractions} = (4 + 2 + 1) / 16 = 7 / 16.$$

$$HM = (3 * 16) / 7 = 48 / 7 = 6.857.$$

Verification Check: $9.333 (AM) > 8.0 (GM) > 6.857 (HM)$. This satisfies the inequality.

CASE PROBLEM 11.4: MULTI-PROCESS VOLATILITY EVALUATION

An operations manager evaluates two production processes over a one-week period. Based on the metrics below, determine which process is more stable and which shows greater asymmetry:

Process X: Mean = 120.0, Median = 115.0, Standard Deviation = 18.0.

Process Y: Mean = 150.0, Median = 148.0, Standard Deviation = 12.0.

Solution:

1. **Evaluate Process Stability (Coefficient of Variation):**

$$CV_X = (18.0 / 120.0) * 100 = 15.0\%.$$

$$CV_Y = (12.0 / 150.0) * 100 = 8.0\%.$$

Process Y is more stable and consistent because it has a lower CV ($8.0\% < 15.0\%$).

2. **Evaluate Asymmetry (Karl Pearson Skewness):**

$$Sk_X = [3 * (120.0 - 115.0)] / 18.0 = 15.0 / 18.0 = +0.833.$$

$$Sk_Y = [3 * (150.0 - 148.0)] / 12.0 = 6.0 / 12.0 = +0.500.$$

Process X shows greater asymmetry because its skewness coefficient is further from zero ($+0.833 > +0.500$).